

Control optimisation of automated guided vehicles in container terminal based on Petri network and dynamic path planning

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ARTICLE INFO

Keywords:

Path optimisation
Obstacle avoidance
Optimisation of vehicle scheduling
Petri network
A-star
Dynamic window approach

ABSTRACT

To solve the problem of path optimisation and obstacle avoidance of automated guided vehicles (AGV) in the horizontal transportation of container terminals, this study proposed three operation modes of terminal vehicles and summarised four vehicle behaviour modes. Considering the vehicle behaviour pattern as the top-level object, based on heuristic rules and an object-oriented timed coloured stochastic Petri network (OOTCSPN) algorithm, a scheduling framework model of the AGV was established to optimise vehicle scheduling. A path planning model for three vehicle operation modes was designed for AGV path planning and real-time random obstacle avoidance based on the A-star(A*) algorithm fusion dynamic window approach (DWA). The simulation results indicate that the container handling efficiency increased by 13.2% using the improved algorithm. The scaled-down experiment confirms that the proposed method makes the planned path smoother and can also achieve real-time obstacle avoidance.

1. Introduction

The objective of horizontal transportation operation in container terminals is to complete container transportation between berths and yards using vehicles for loading and unloading of containers. With the increase in global trade volume and development of large-scale ships, manual operation of container trucks has faced problems such as recruitment difficulties, high wages of workers, frequent accidents, and weak accident response ability that cannot meet the requirements of cost reduction, efficiency increase, and service quality improvement. An increasing number of container port companies and researchers are exploring automated container handling methods.

In terms of intelligent driving, Chang'an University has successively set up test bases to conduct autonomous driving tests for vehicle-road coordination [1]. However, it focuses more on the automation of a single vehicle. Some related studies have focused on the analysis and optimisation of traffic flow characteristics of the AGV fleet [2], vehicle motion parameter simulation [3], and vehicle control [4–5]. These studies primarily focus on theoretical research and simulation tests in the field of transportation and are difficult to be verified practically. Li [6] proposed a lane-change decision-making framework based on deep reinforcement learning to determine a risk-aware driving decision strategy with the minimum expected risk for autonomous driving. However, application verification has not been conducted for horizontal transportation operations in container terminals. In terms of vehicle fleet scheduling, some researchers have focused on optimisation of vehicle task allocation and maximisation of vehicle efficiency or utilisation

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rate [7–10]. Others have primarily studied the integrated scheduling problem between the AGV and equipment [11–12] to better coordinate the work between them; however, most researchers have simplified the driving path of the AGV during the test. In terms of AGV path planning, researchers have focused on path optimisation or vehicle conflict problems, but the constraints have been simplified or idealised. Zhong [13] realised path planning and integrated scheduling between AGV and lifting equipment based on a mixed-integer programming mode. However, they did not consider the problem of random obstacle avoidance during vehicle driving. Zhang et al. [14] proposed a deep reinforcement learning-based hyper-heuristic model to optimise the container truck routing problem by considering uncertainties in service times. Ghasemzadeh [15] proposed a multi-AGV path-search method to avoid road congestion based on a static network graph. Gawrilow et al. [16] considered the load balance and recessive time of automated guided vehicles (AGVs) and adopted the graph theory method to plan conflict-free AGV paths. In summary, in terms of research on horizontal transportation operation management and control in container terminals, little comprehensive research exists on the scheduling and path planning of AGVs, or on the conflicts between vehicles and obstacles. Even if the conflict is considered, few effective real-time prevention or solution measures exist.

Considering these problems, this study proposes three innovative AGV running modes and designs an AGV scheduling model. Based on the A* algorithm fusion DWA and the feedback of a sensor, the collaborative perception and dynamic path planning of AGVs are realised. Independent handling of the containers was achieved.

The main contribution of this work is summarised as follows.

A scheduling framework model of the AGV fleet is proposed in this study based on heuristic rules and OOTCSPN. The proposed model can direct the traffic order and position sequence of an AGV fleet.

In the process of vehicle driving, we calculated the global critical path points based on the improved A* algorithm to direct the vehicle drive to the destination. Other moving AGVs and objects entering by mistake are random obstacles that can easily cause collision accidents. To solve these problems, DWA is used for real-time path planning between critical path points.

The proposed method can improve the operation efficiency of the AGV fleet and achieve random obstacle avoidance. The results demonstrate that the proposed scheme performs well in terms of task scheduling and path planning.

The remainder of this paper is organised as follows. Section 2 summarises the operation mode and characteristics of AGV in container terminals and highlights the problem. Section 3 provides the details of the proposed method. The experimental results and analysis are presented in Section 4. Section 5 concludes the study.

2. Design of Horizontal Transport Operation Mode

Container terminals have both AGVs and traditional container trucks for serial/parallel operations such as charging, parking, and transportation. There are hundreds of vehicles at the peak of operation. Intelligent construction of a horizontal transportation system should be carried out according to the layout and size of the terminal to control the orderly operation of vehicles. The following three vehicle operation modes were designed in this study, as shown in Fig. 1.

The fully isolated mode can ensure no crossover between AGVs and trucks, and has the advantages of relatively simple control and no risk of collision between AGVs and trucks. However, the utilisation rate of roads is relatively low; thus, they are suitable for terminals with broad sites and weak AGV control ability. In semi-isolated mode, there is a collision risk at the intersection of AGVs and trucks. Thus, the requirements for the control ability of AGVs have been improved. The road utilisation rate can be improved and is suitable for docks with strong AGV scheduling and control capabilities. In the hybrid autonomous operation mode, there is no need to divide the vehicles and their driving areas, and all vehicles can realise autonomous cross-operation, which is the ideal operation mode for horizontal transportation operations of the terminal. It has the advantages of high intelligence, high road utilisation rate, perfect control, and scheduling strategy. Owing to the lack of research and practice on vehicle road coordination, a hybrid autonomous operation mode has not yet been implemented in existing terminals. Regardless of the vehicle operation mode, if the scheduling control

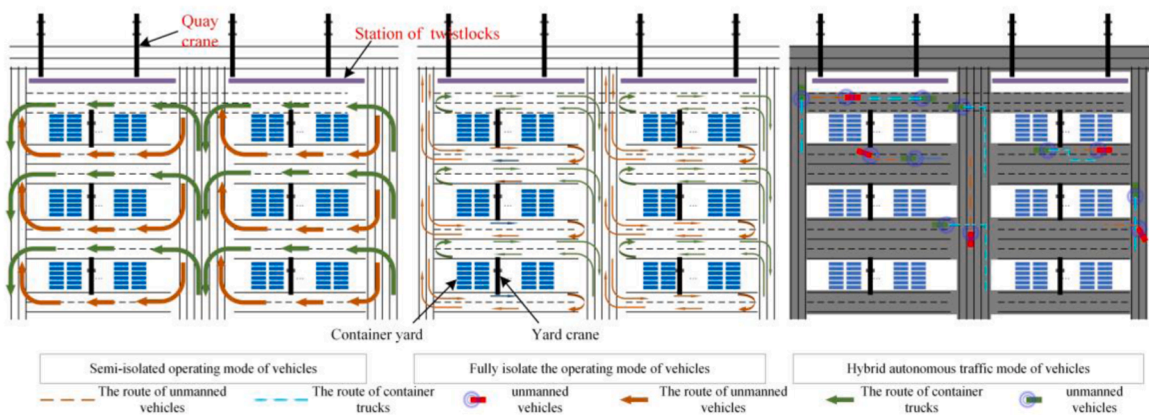


Fig. 1. Operation mode of vehicle in terminal

and obstacle avoidance optimisation of the AGV fleet are not carried out, risks such as collisions between AGVs or static/dynamic obstacles exist. Thus, task optimisation and path planning of vehicles considering dynamic obstacle avoidance are urgent problems to be solved.

3. Modeling and Algorithm Design of AGVs

The loading and unloading operations of the terminal must be handled by AGVs. For the loading operation, the parking space under the quay bridge is the starting point of the transfer task, and the parking space in the yard is the end point. For unloading operations, the starting and ending points are opposite. Various sensors on the vehicle can feed back environmental information in real time during the entire process. The dispatching control software carries out all vehicle dispatching control and real-time planning to ensure that the AGV can safely reach the destination. The flow of the horizontal transport operation is shown in Fig. 2.

3.1. Scheduling and Control Modelling

There are five behaviour modes of vehicles in the terminal: charging, parking, driving, disassembly, and assembly of the twist lock. Moreover, there are crossover operations of vehicles in the terminal; thus, it is easy to cause collisions and other accidents. The dispatch order, operation time, and conflict occurrence of AGVs are random; therefore, the horizontal transportation operation belongs to a discrete system. The Petri network, proposed by Dr. Carl Adam Petri, is widely used in software engineering and manufacturing system modelling [17]. In addition, it has been applied to research on logistics systems by many scholars [18–20]. From the perspective of the Petri network model, the objective of horizontal transportation operation management is to confirm the optimal operation sequence of the AGV and to solve the conflict problem in the driving process. There were tasks with multiple vehicles and links in parallel. To enhance the description ability of Petri networks for complex systems, this study established the OOTCSPN [21] model of horizontal transportation operations by combining object-oriented modelling technology and heuristic rules. Finally, the system information in terms of architecture and state transition was obtained. The OOTCSPN modelling process can be divided into three parts. The first part is the PN model of the top-level object. In this study, the four behaviour modes of vehicles, which are top-level objects, are considered as the research object. The second part is the Petri model of a single object. This layer adopts object-oriented technology to encapsulate the main objects used in the system and describe their specific operation processes. The third part is the door transition between the top-level objects.

The stochastic Petri network of the object-oriented assignment for the scheduling and control of AGVs in the terminal can be described by N , which is defined as a triplet. The Petri network model is shown in Eq. (1).

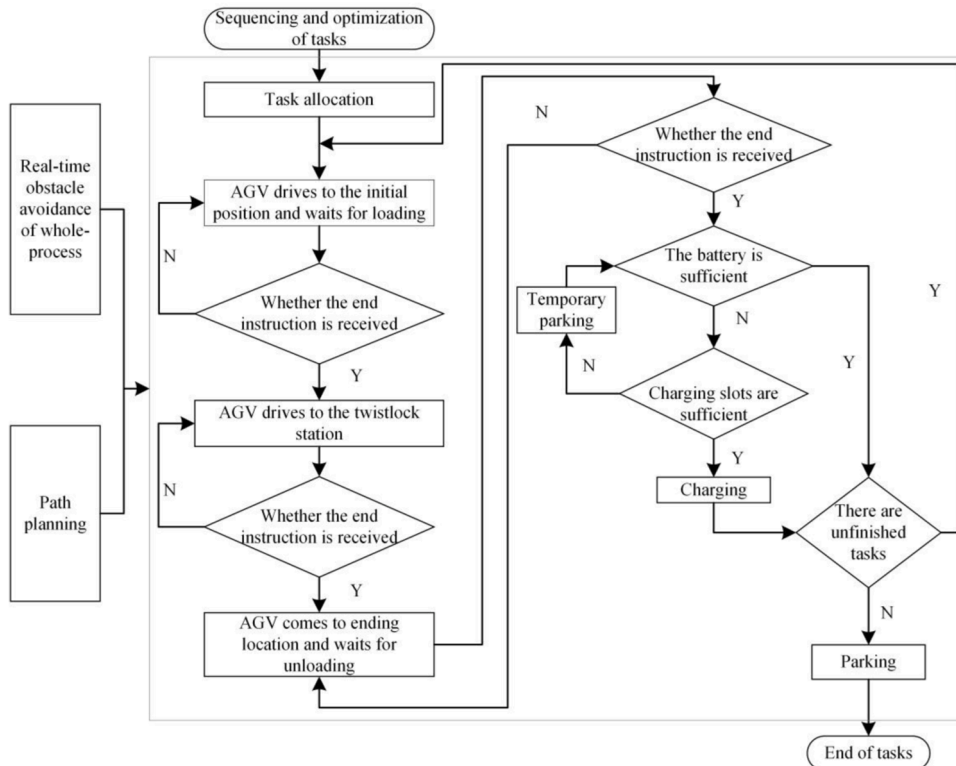


Fig. 2. Horizontal transportation operation in terminal

$$N = \{Ob_i, G_{ij}, M_0\} \quad (1)$$

where $Ob_i = \{SP_i, AT_i, IM_i, OM_i, I_i, O_i, H_i, D_i, A_i\}$ represents the i^{th} object of the system, referring to the four behaviour modes of the AGV herein;

SP_i represents a finite set of places p of the i^{th} object, indicating different states of various resources. In this model, a place contains resources and information bits. The token in resource bits represents the state of objects or resources used or transported in the system, and its flow in Petri nets represents the change in material flow or resource state. The token in the information bits represents the information transmitted in the system, and its flow describes the transmission of information flow.

AT_i represents a finite set of transition T of the i^{th} object, which contains operational procedures or instructions that cause resource state changes.

IM_i represents a finite place set of input information on the i^{th} object;

OM_i represents a finite place set of output information on the i^{th} object;

I_i represents the duplicate dataset of directed arcs from P to T of the i^{th} object, which is a set of non-negative integers.

O_i represents the duplicate dataset of directed arcs from T to P of the i^{th} object, which is a set of non-negative integers.

H_i represents the colour set of the place and transition for the i^{th} object. The complexity of this system has increased due to the large number and variety of resources. Thus, colouring was introduced to aggregate and classify resources according to their functions. In this model, resources are divided into operation rules (mainly including scheduling rules), equipment classes (mainly including AGVs, charging piles, twist-lock, lifting equipment), and location classes (mainly including shore parking spaces, twist-lock stations, charging spaces, yard parking, and parking spaces).

D_i represents the time set of the place and transition for the i^{th} object. In this study, once a transition is triggered, a certain number of tokens are immediately subtracted from the input place of the transition. With operations such as container handling, loading, and unloading, AGV behaviours require a certain amount of time to be completed. The transition can only be stimulated with a certain time delay; a corresponding number of token numbers will be added to the output place.

A_i is an excitation function that relates the excitation rate to the transition of the i^{th} object. The excitation function was used to model discrete dynamic events with uncertain time. This model primarily relies on heuristic rules and process flows.

G_{ij} represents the information transfer network from object ob_i to object ob_j , and M_0 denotes the initial identity of the system.

3.1.1. Scheduling Strategy and Modelling of Vehicle Charging

The design of charging module OOTCSPN model is shown in Fig. 3.

It is observed in the reachability graph that the limitation of charging stations and the diversity of AGV states lead to conflicts regarding whether the vehicle can be charged. This problem can be solved based on the following rules, which are sorted from highest to lowest priority.

- (1) When the electric quantity of the vehicle filled with objects is less than p_1 , it arrives at the specified position to charge after completing the current task.
- (2) When the average electric quantity of all vehicles filled with objects is less than p_3 , all idle vehicles with a charge higher than p_4 are preferentially queried, and the least-charged vehicle after unloading is in turn replaced by idle vehicles with more power.

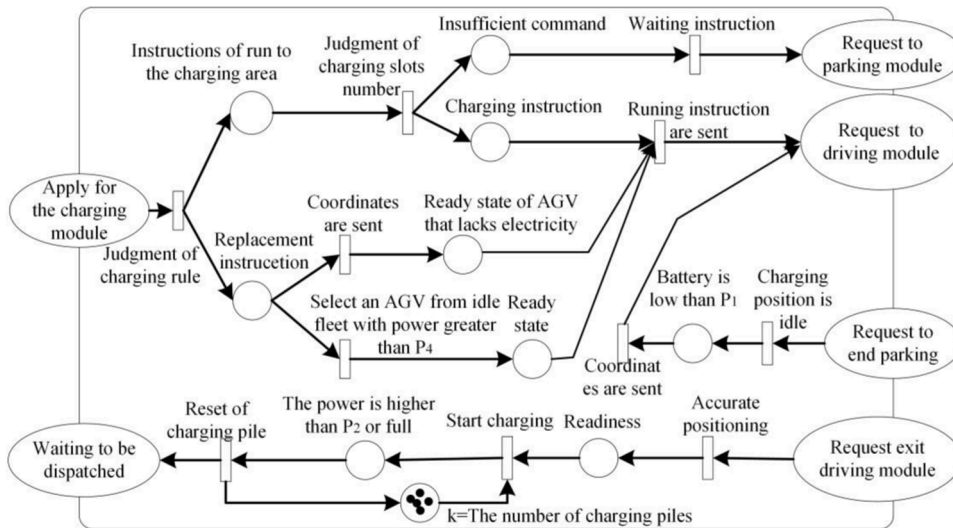


Fig. 3. OOTCSPN model of charging module

- (3) When the electric capacity of the vehicle in charge reaches 100%, it applies online and waits for scheduling. The system forces the vehicle with the lowest power to charge offline after completing the current task before returning online when it is fully charged.
- (4) When the charging vehicle's power is higher than p_2 , the AGV reports the application to perform the task at any time.

3.1.2. Scheduling Strategy and Modelling of Vehicle Parking

The design of the parking module OOTCSPN model is shown in Fig. 4.

The parking control strategies of AGVs are as follows.

- (1) When the current task is completed, if there is no charging pile and the power of the AGV is less than the threshold value, the AGV will first complete temporary parking. The driverless vehicle is then in the charging state and will be charged actively when there is an idle charging position.
- (2) At the end of the current task, if the power meets the requirements and no task is performed, the AGV will be set to the idle state to complete the parking operation.

3.1.3. Scheduling Strategy and Modelling of Vehicle Driving

The OOTCSPN model of the driving module of AGVs includes five sub-objects: the module to destination, the module to origin, scheduling module, vehicle running module, and conflict resolution module, as shown in Fig. 5.

It is observed in the reachability graph that there are collisions between AGVs and static/dynamic objects at the intersection if they are not controlled. This problem can be solved using a heuristic scheduling strategy for vehicles. The following rules were sorted from high to low:

- (1) The AGV passing level is higher than that of traditional container trucks.
- (2) When the waiting time of a container truck exceeds p_5 , or there are more than two container trucks in an area simultaneously, the AGV traffic level is reduced, and container trucks are given priority.
- (3) The same type of car shall follow the principle of first-come first-served.
- (4) For vehicles of the same type, those with high task priority should be given priority.
- (5) For vehicles of the same type, vehicles that take a long time are prioritised.

3.1.4. Scheduling Strategy and Modelling during Disassembly and Twist-lock

The design of the OOTCSPN model for the twist-lock module is shown in Fig. 6.

It is observed in the reachability graph that if the twist lock equipment cannot be configured reasonably, there is congestion or collision between AGVs. This problem can be solved using the following heuristic scheduling strategy; the following rules are sorted in descending order of priority:

- (1) Vehicles closer to the twist lock station have higher priority.
- (2) The strategy of maximising the effective utilisation of unlocking stations is used to allocate the corresponding twist lock equipment to the vehicles.

The OOTCSPN model of horizontal transportation operation is shown in Fig. 7.

3.2. Algorithm Design of Path Planning

AGVs can perceive multisensor information and provide basic automatic control. However, they have no provision for direct information sharing or intelligent management between vehicles in the fleet, which makes them vulnerable to collisions and other accidents. Thus, it is necessary to develop a scheduling algorithm and software to control the vehicle fleet. Path planning refers to the

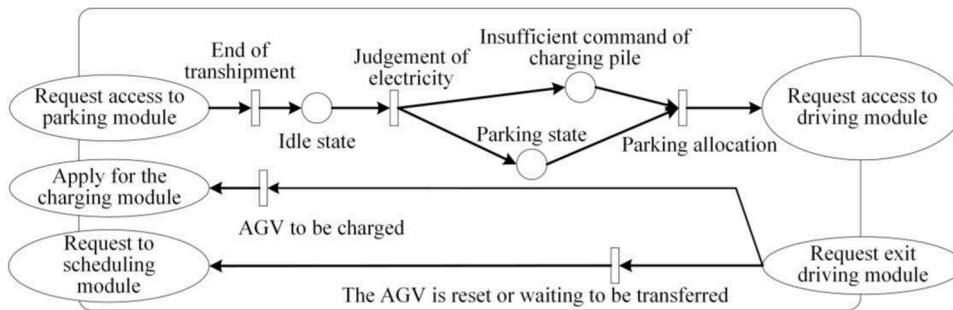


Fig. 4. OOTCSPN model of parking module

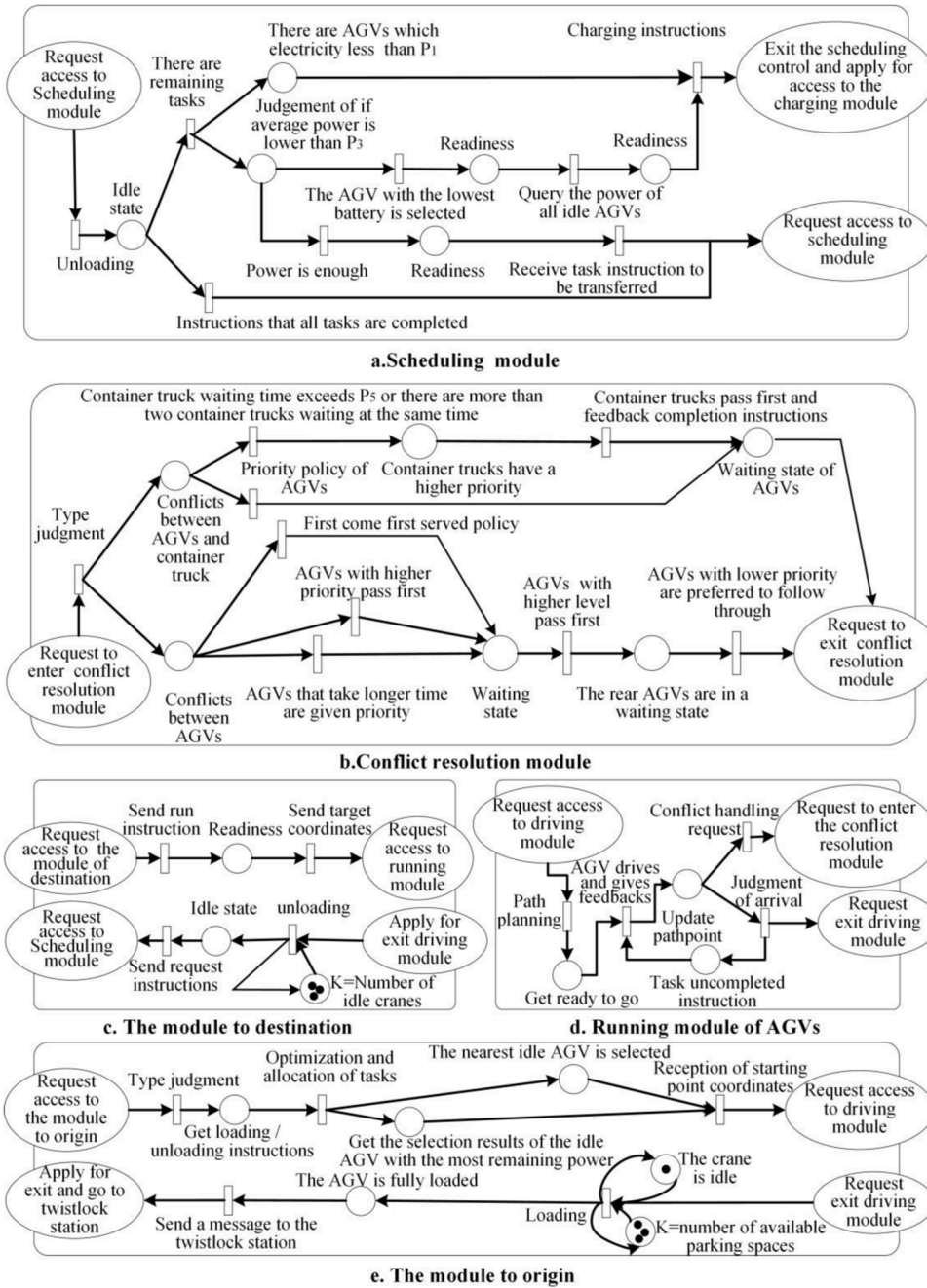


Fig. 5. OOTCSPN model of vehicle running module

automatic planning of a collision-free path from the starting point to the end point by combining an environmental map and sensor perception feedback information [22]. In this study, global planning and dynamic local planning methods were used for real-time AGV path planning.

The A* algorithm is a global path planning method [23], an algorithm for finding the lowest cost for a path with multiple nodes on the plane. The global path points of AGVs in the container terminal can be generated based on the A* algorithm. By estimating the $h(i)$, this study can reduce the possibility of detours and find an excellent path. The A* algorithm is expressed in Eq. (2).

$$f(i) = g(i) + h(i) \quad (2)$$

where $f(i)$ is the estimated cost, $g(i)$ is the actual cost from the starting point to the child node i , and $h(i)$ is the estimated cost for the child node i to the target point. Among them, $g(i)$ is the product of the cost of movement and the cost factor, and there are several ways

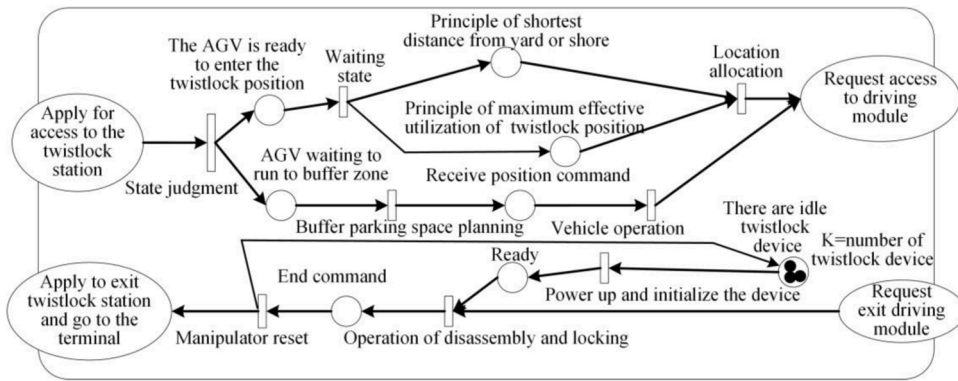


Fig. 6. OOTCSPN model about twist-lock

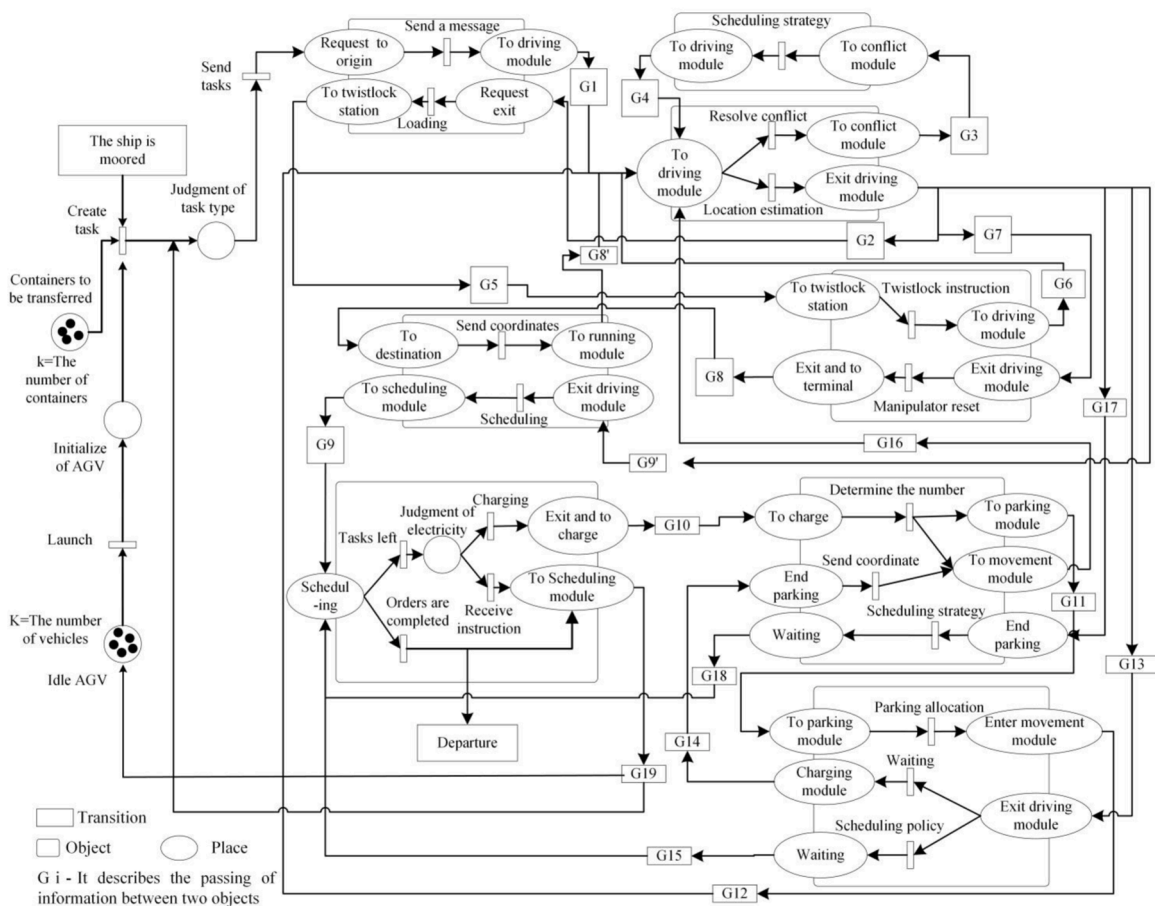


Fig. 7. OOTCSPN model of cooperative vehicle-infrastructure system

to estimate $h(i)$, such as the Manhattan distance, Euclidean distance, and diagonal estimation.

To improve the search speed for the optimal global path points of the A* algorithm, this study optimises the search point selection strategy. The target point is connected to the current point; the included angle between the line and the N_1 direction was set as δ , and the three search directions of the back pair were discarded. The optimisation process for point selection is shown in Fig. 8.

In this study, the backward-facing obstacle search method [24] was used to overcome the problem of the path planned by the traditional A* algorithm being too close to obstacles. The path generated by the A* algorithm is not sufficiently smooth and lacks dynamic obstacle avoidance capability. Thus, the critical global path points based on the A* algorithm are retained. When the AGV is driving to a global key point P_c , if the distance d_i between the nearest random obstacle and P_c is less than the safety distance D , the key

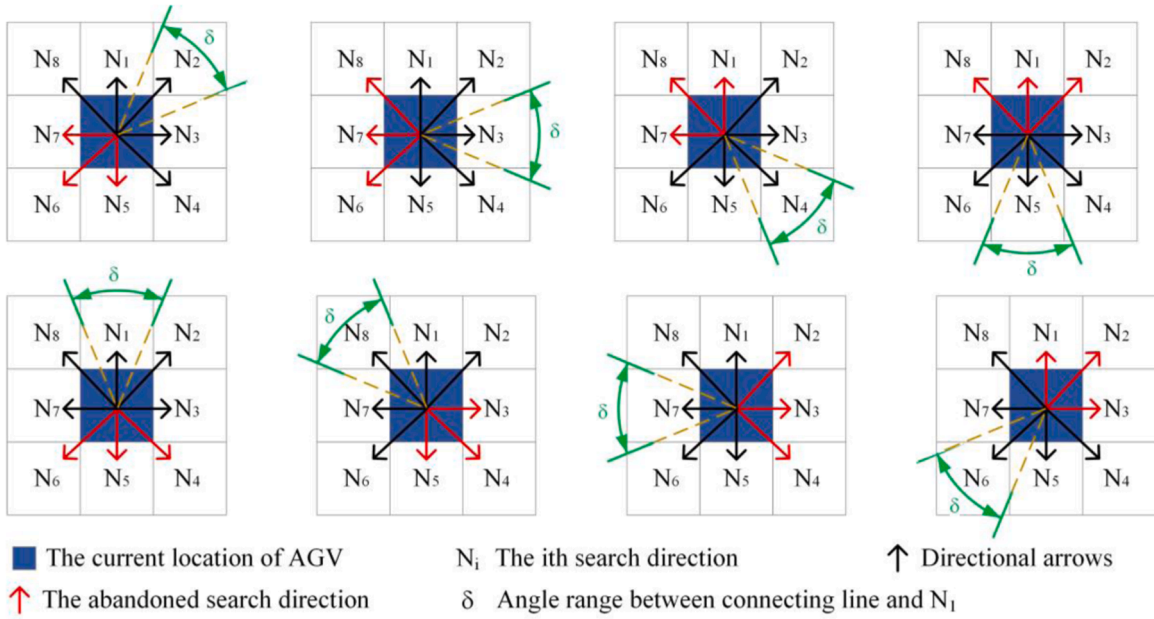


Fig. 8. Point selection optimisation process

global path point is updated as follows. The current global key point P_c is replaced by P_c' . The position of the current key point P_c' is updated with P_c as the centre, D as the radius, and in the direction close to the next key point P_n . The keypoint update process is illustrated in Fig. 9.

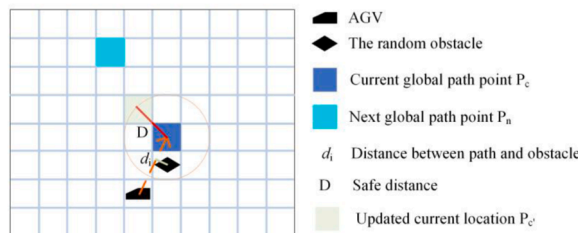
The DWA [25] is used between two adjacent critical path points for path local optimisation and random obstacle avoidance. Let $v(t)$ and $w(t)$ respectively represent the velocity and angular velocity of the vehicles at time t in the world coordinate system. Within the sampling period Δt , the displacement can be approximated as a linear motion, and its kinematics model can be expressed as Eq. (3).

$$\begin{aligned} x(t) &= x(t-1) + v(t)\Delta t \cos(\theta(t-1)) \\ y(t) &= y(t-1) + v(t)\Delta t \sin(\theta(t-1)) \\ \theta(t) &= \theta(t-1) + w(t)\Delta t \end{aligned} \quad (3)$$

The AGV is affected by its own motion parameters, motor performance, and braking distance during operation; the aforementioned constraints have a certain control range, as shown in Eq. (4).

$$\begin{aligned} V_m &= \{(v, w) | v \in [v_{\min}, v_{\max}], w \in [w_{\min}, w_{\max}]\} \\ V_d &= \{(v, w) | v \in [v_c - a_{d\max}\Delta t, v_c + a_{i\max}\Delta t], w \in [w_c - \alpha_{d\max}\Delta t, w_c - \alpha_{d\max}\Delta t]\} \\ V_l &= \{(v, w) | v < \sqrt{2dist(v, w)a_{d\max}}, w < \sqrt{2dist(v, w)\alpha_{d\max}} \end{aligned} \quad (4)$$

where V_m represents the speed constraint of the vehicle, V_d denotes the speed constraint of the motor under acceleration and deceleration, and V_l indicates the speed range of the AGV considering the safety constraints. $v_{\max}$ and $v_{\min}$ refer to the maximum and minimum linear velocity, respectively; $w_{\max}$ and $w_{\min}$ represent the maximum and minimum angular velocities, respectively; v_c and w_c denote the linear velocity and angular velocity at the current moment, respectively; $a_{d\max}$ and $a_{i\max}$ represent the maximum deceleration, respectively; $\alpha_{d\max}$ and $\alpha_{i\max}$ indicate the maximum acceleration, respectively; A and D refer to the maximum angular deceleration and maximum angular acceleration, respectively; $dist(v, w)$ represents the shortest distance from the end of the simulated trajectory to the obstacle.

Fig. 9. Global key point updating method of A* algorithm when $d_i < D$

The final speed of the DWA is the intersection of these three constraints; the speed V_r is expressed as Eq. (5).

$$V_r = V_m \cap V_d \cap V_l \quad (5)$$

After the AGV trajectories are obtained, the evaluation function is used to score these trajectories to screen out the optimal trajectory.

$$G(v, w) = \alpha \text{Head}(v, w) + \beta \text{Vel}(v, w) + \gamma \text{Dist}(v, w) \quad (6)$$

Herein, α, β and γ are the weighting coefficients; $\text{Head}(v, w)$ is the direction angle deviation between the end direction of the trajectory and the target point; $\text{Vel}(v, w)$ is used to evaluate the speed of the AGV running in the current trajectory; $\text{Dist}(v, w)$ refers to the shortest distance between the end point of the simulated trajectory and the obstacles.

Based on the feedback of the environmental information, scheduling strategy, and AGV path planning algorithm, the AGVs can avoid moving/static obstacles while travelling and simultaneously ensure the shortest path and realise path optimisation. The scheduling optimisation process is illustrated in Fig. 10.

4. Experimental Design and Validation

4.1. Simulation Verification

A raster map of the container terminal is designed, which includes the ship docking area, quayside operation area, yard area, and vehicle driving area. The codes for each area are shown in Fig. 11. The process and location coordinates of the AGV are assigned using a scheduling framework based on heuristic rules and OOTCSPN. The traditional A* algorithm and improved path planning algorithm were used to simulate on-site vehicle management and control. In the simulation experiment, $\triangle$ indicates the starting point, $\circ$ indicates the endpoint, and a rectangle indicates an AGV. It is assumed that there are six AGVs and the terrain is flat. The estimated actual cost of the four motion directions of the vehicle is set to 10. The actual cost of diagonal movement of the vehicle is set to 14. The estimated cost is calculated using the Manhattan distance method. There were 10 tasks in this experiment. The location code for the task is presented in Table 1. The two tasks with location codes (A5 and C3) and (A1 and B6) have relatively long transportation distances.

The DWA parameters are set as follows: The maximum speed is 1 m/s, the maximum angular speed is $10^\circ/\text{s}$, the speed resolution is 0.01 m/s, the angular speed resolution is $2^\circ/\text{s}^2$, the acceleration is 0.1 m/s^2 , and the angular acceleration is $2^\circ/\text{s}^2$. The parameter values of the evaluation function are $\alpha=0.1, \beta=0.05$, and $\gamma=0.2$. The prediction time was set as 1 s. As shown in Fig. 11, when there are

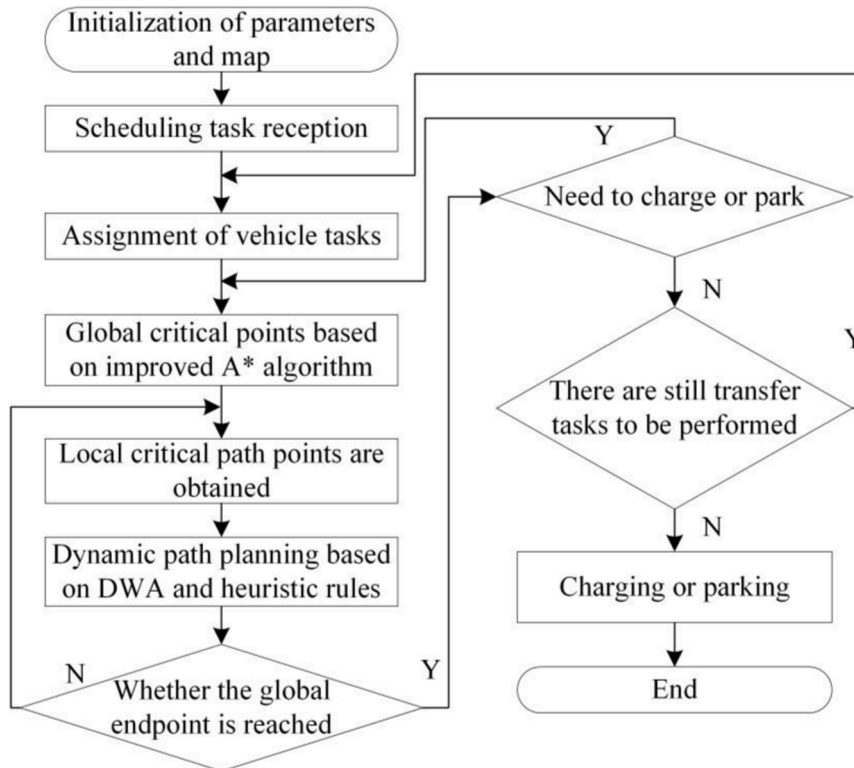


Fig. 10. Flow chart of vehicle-road collaborative control

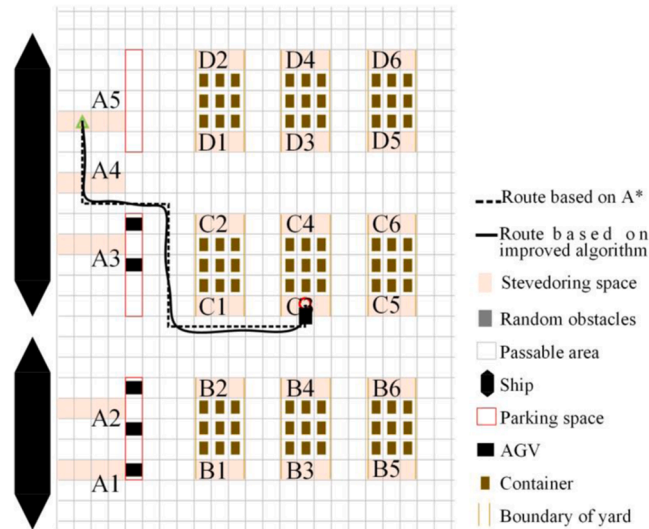


Fig. 11. Map layout and planning results with two algorithms

Table 1
Task location codes

Task	Vehicle 1	Vehicle 2	Vehicle 3	Vehicle 4	Vehicle 5	Vehicle 6
1	(A5, C3)	(A1, B6)	(B3, B2)	(C2, A3)	(B2, A2)	(A3, C5)
2	(C3, A1)		(B2, A3)	(A3, D3)	(A2, B1)	

no random obstacles in the map, both A* and the improved algorithm can plan an optimal path from the starting point to the target point. However, the path planned by the improved algorithm was smoother than that planned by the A* algorithm.

As shown in Figs. 12 and 13, the AGVs based on the A* algorithm cannot avoid sudden obstacles, whereas the AGVs based on the improved algorithm can achieve random obstacle avoidance. In Figs. 14 and 15, the speed of AGVs based on the A* algorithm varies greatly, and the probability of the speed being 0 is high because AGVs cannot avoid random obstacles under the control of A*. When the radar of an AGV detects random obstacles and the safety distance exceeds the standard, the AGV brakes urgently. However, the improved algorithm avoids these problems.

In Table 2, the path planned by the improved algorithm may be longer with an increase in the number of obstacles. As there is no time wastage caused by vehicle emergency braking and restart, the overall planning efficiency of the improved algorithm is relatively

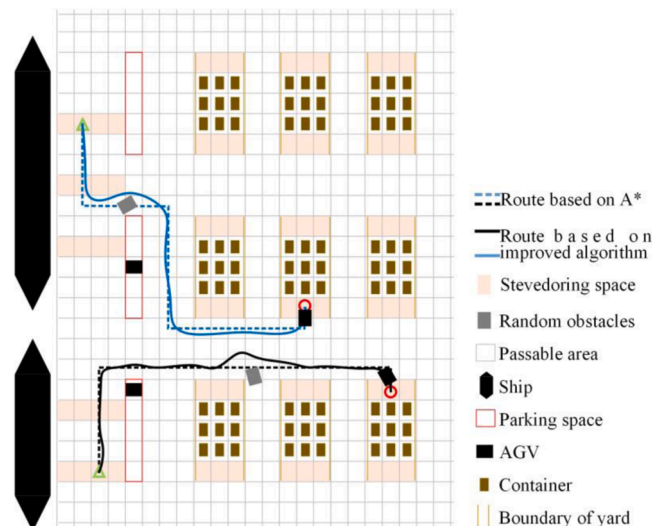


Fig. 12. Path planning with one random obstacle

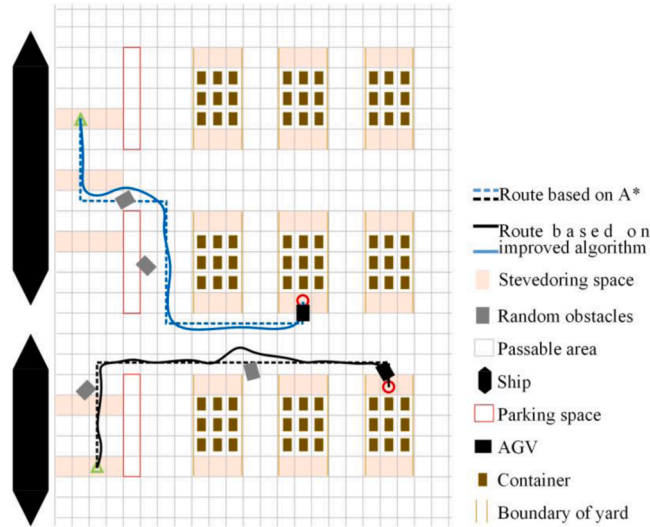


Fig. 13. Path planning with two random obstacles

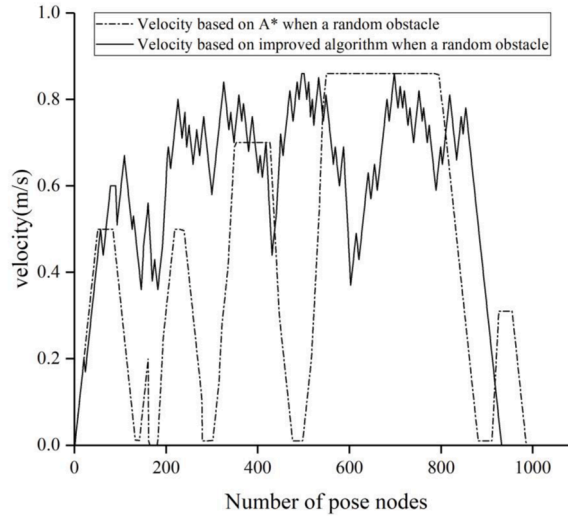


Fig. 14. Velocity curve with one obstacle

high.

In this experiment, the vehicle based on A* has four emergency braking cycles, whereas the vehicle based on the improved algorithm does not have this problem. The task based on A* required 174 s; the task based on the improved algorithm required 151 s. The efficiency of the improved algorithm was increased by 13.2%. The results are shown in Figs. 16 and 17.

From the task Gantt in Fig. 17, this scheduling framework can effectively command the AGV fleet to complete the container handling tasks. During the same period, vehicles with high priority can be prioritised in the intersection area. AGV1 moves from A5 to C3. AGV1 intersects with AGV6 while driving. AGV1 has a higher priority due to its longer path. AGV6 brakes first and starts driving again after AGV1 passes. AGV2 drives from A1 to B6. AGV2 and AGV3 meet in this manner. AGV2 has a higher priority owing to its longer path. When the two AGVs meet, the speed of AGV3 is 0. The results are shown in Fig. 18. The experimental results indicate that the scheduling framework based on OOTCSPN can effectively command traffic order and avoid collisions in the intersection area.

4.2. Scaled-down Experimental Verification

A scaled-down validation laboratory was established. One quayside crane, two automated rail-mounted container gantry cranes, one container ship, three AGVs, several container models, and one centralised control centre were deployed in the laboratory. The site area was divided into the quayside operation area, AGV buffer area, AGV driving road area, yard area, charging area, and central

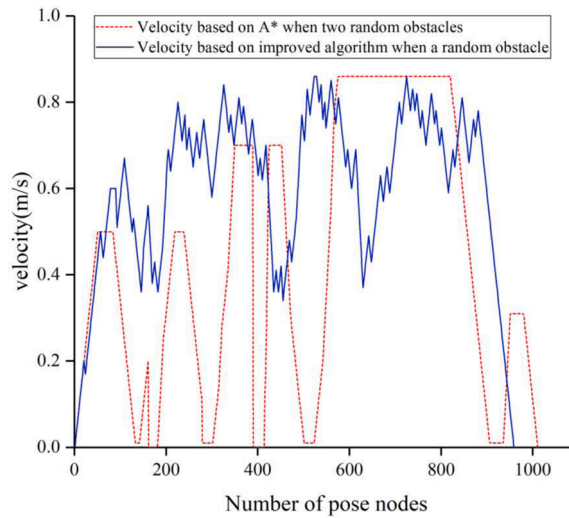


Fig. 15. Velocity curve with two obstacles

Table 2
Path planning performance comparison of one vehicle

Number of random obstacles	Algorithm	Path length (m)	Total time (s)
1	A*	30	95
	Improved algorithm	34	91
2	A*	31	106
	Improved algorithm	36	98

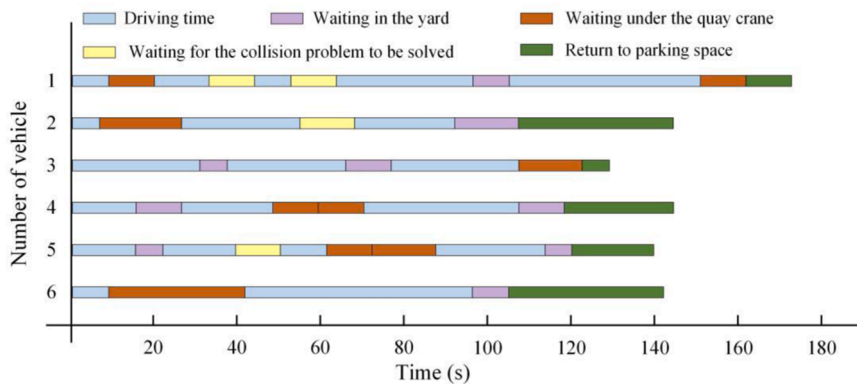


Fig. 16. Time of all tasks based on A* algorithm

control centre area. An on-site local area network (LAN) was constructed. The equipment communication was based on the LAN. AGVs interact with scheduling software through wireless fidelity. The AGVs navigate through laser radar and millimetre-wave mines. With the aid of an alignment tripod, parking positioning accuracy can reach 2cm. The sampling period was 0.1s. The scanning range was 0–18 m. The maximum speed was 0.8m/s. The maximum angular velocity was 0.08 rad/s. The safety distance of the AGV was set as 0.8 m. A map of the laboratory environment was constructed by moving the AGV manually. A plan and map of the laboratory environment are shown in Fig. 19.

The red scattered dots in the map represent the real-time observation data collected by LIDAR; the black scattered dots represent the boundary areas, such as walls and baffles in the laboratory; the light green rectangular area is the driving area of the laboratory, and the blue rectangles are the collision zones on the roads where vehicles are travelling. Non-driving areas (including static objects in the area) are considered as static obstacles. The moving vehicles were movable obstacles used during the experiment. The participants entering the driving area during the experiment included random obstacles. The starting and ending points of the AGVs included the loading and unloading positions on the shore or yard, respectively. Based on the scheduling strategy and path planning algorithm, the AGVs can arrive at the destination and assist the lifting prototype in completing the container loading and unloading tasks. The

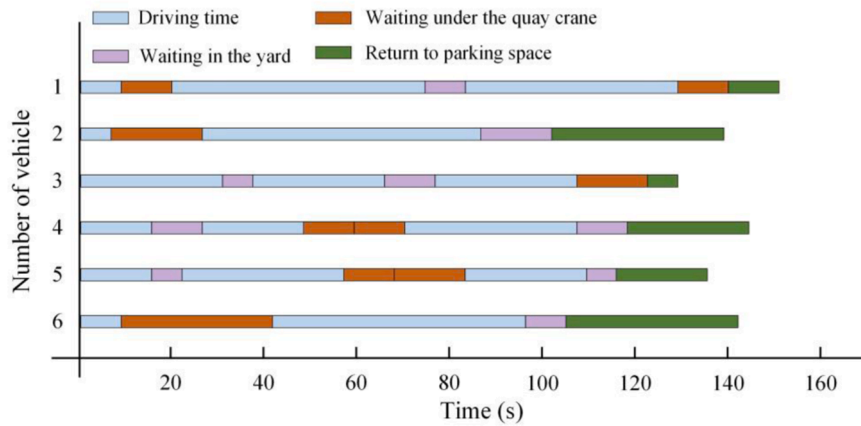


Fig. 17. Time of all tasks based on improved algorithm

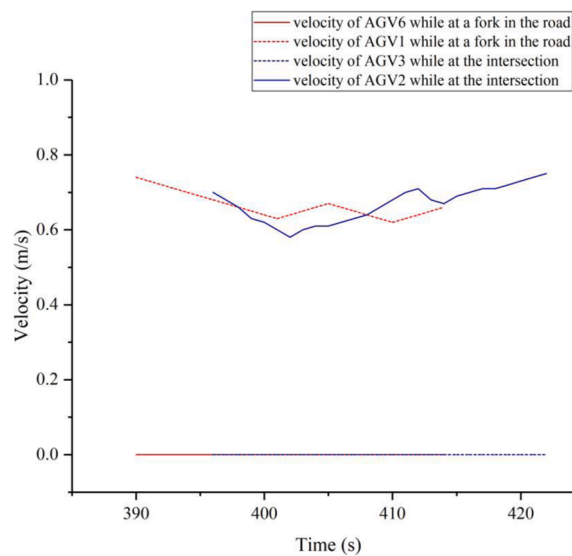


Fig. 18. Diagram of vehicle scheduling results at intersection in same time period

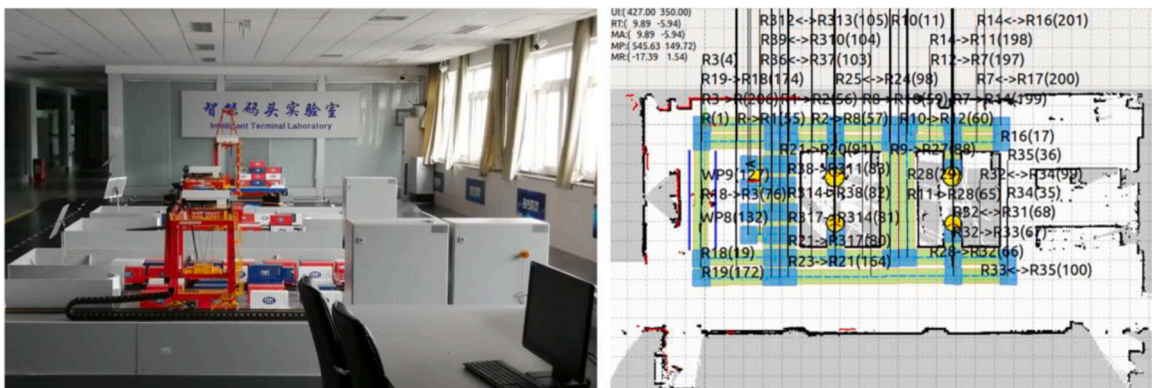


Fig. 19. Layout and map of intelligent terminal laboratory

Methodology, Funding acquisition. **Le Li**: Formal analysis, Funding acquisition. **Yue Wang**: Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data Availability

The authors do not have permission to share data.

Acknowledgement

This work was funded by the National Natural Science Foundation of China under Grant Nos. 51979228 and 61903304.

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